

A Snapshot of Economics

Mapping The New Palgrave Dictionary and its Evolution

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Fragmentation or Homogeneity?

- Different literary genres provide opposite images of the current state of the discipline:
- **research frontier** (journals, specialized handbooks)
 - *Death of Neoclassical Economics* (Colander 2000)
 - *Mainstream Pluralism* (Davis 2006)
 - *Age of Fragmentation* (Roncaglia 2019)
 - *Disunity of Economics* (Ambrosino et al 2021)
- **education** (introductory textbooks, syllabus) find remarkable homogeneity and stability over time and space
 - textbooks (Bowles & Carlin 2020)
 - graduate programs (Fourcade et al. 2015)

What about other sources and genres?

Encyclopedias as disciplinary snapshots

- Encyclopedias are an **underexplored source** in the historiography of economic thought (Besomi 2012).
- Unlike research articles, they present **stable knowledge**, not cutting-edge theories.
- Their role is both **descriptive and normative**:
 - Inclusion signals what counts as reliable knowledge.
 - Absence reveals what was seen as irrelevant or marginal.
- Like textbooks and handbooks, they **codify** the discipline:
 - Shape what students learn and what scholars consult.
 - Contribute to the construction of disciplinary boundaries.

The breadth of coverage of New Palgrave renders impossible any attempt to comment upon its contents in detail, for to do so would involve the whole corpus of economics, past and present.

(Bailey, 1994)

The Palgrave Dictionary - Early History

- **1894-1899:** First complete British encyclopedia focused on economic theory, edited by R. Palgrave.
 - Didactical mission: serve students, present established knowledge and biographical entries.
 - Strongly **British in content and authorship:** Scots law, colonies, trade tables, UK universities.
 - Featured a range of views, with marginalism as analytical backbone.
 - Notable contributor: **F. Y. Edgeworth** (100+ entries).
- **1925-26:** Revised edition under H. Higgs.
- No similarly ambitious project for decades.

The New Palgrave - Recent Evolutions

- **1987:** edited by Eatwell, Milgate, Newman.
 - Reflects modern economics: econometrics, game theory.
 - Historical framing requested in every entry; HET and biographical content remains key.
 - Reprints from original Palgrave juxtaposed with new entries (e.g. Edgeworth vs Debreu on Mathematical economics).
 - Strong emphasis on heterodox perspectives.
- **2008:** Fully revised, edited by Blume and Durlauf.
- **Living** (online) and incoming reworking edited by Vernengo, Perez Caldentey, Ghosh.
- Most prestigious: 41 Nobel winners contributed (+4 since we started!).

Too Heterodox?

hours of labour. *See* EXPLOITATION.

- Stigler (1988) and Blaug (1988) praised The New Palgrave, but criticized the excessive space devoted to heterodox perspectives (especially Marxian and Sraffian).
- Milgate (1992) explains the production process and defends editorial choices and epistemic project.
 - How much is too much?
 - Who wrote the entries?
- What happens in 2008? Why the silence?

- **Source:** web scraping of the official online Living edition.
- **Informations for each entry:** DOI, Title, Authors, Text, Abstract, Bibliography, Date, JEL codes (only for 2008+).
- Living edition features a total of 3119 entries:
 - 284 entries published online since 2009;
 - 1141 entries newly published or significantly updated for the 2008 edition;
 - 709 entries published in 1987 which got reprinted in 2008;
 - 983 entries published in 1987 that did not get incorporated in 2008.

This allows to reconstruct the entirety of the 2008 edition, but we lose some entries of the 1987 one.

Combining Text and Bibliography

- We use **Similarity Network Fusion** to integrate:
 - Text-based similarity;
 - Bibliographic similarity.
- **How it works:**
 - Construct similarity matrices from each data source.
 - Iteratively update each network by blending it with the others: at each step, nodes “share” similarity information with their neighbors across all networks.
 - This process reinforces shared patterns while reducing noise. Preserves the structure of each modality while reinforcing shared signals.
- Result: a network of entries with links based on both **thematic content** and **referential proximity**.

Layer1: Text

Normalized Cosine Similarity on Text Embeddings

- Each entry's abstract is embedded using **Sentence-BERT**:

$$\mathbf{e}_i = \text{SBERT}(\text{Abstract}_i)$$

- Compute pairwise **cosine similarity**:

$$\cos(i, j) = \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\| \|\mathbf{e}_j\|} \in [-1, 1]$$

- Normalize** to adjust scaling and ensure bounded, interpretable similarity:

$$\tilde{S}_{ij} = 1 - \frac{\sqrt{2(1 - \cos(i, j))}}{2} \in [0, 1]$$

Layer 2: Bibliography

Jaccard Similarity on Bibliography sets

- **Jaccard Similarity:**

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \in [0, 1]$$

where A and B are the sets of references in two entries.

- Captures the degree of overlap between bibliographies, **independent of length.**

Hierarchical Clustering

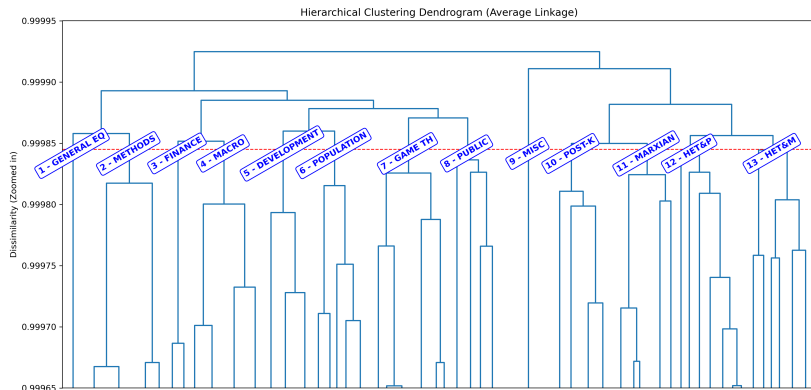
- We apply **agglomerative hierarchical clustering** to the fused similarity matrix:
 - Start with each entry as its own cluster.
 - Iteratively merge the two closest clusters.
- Closeness between clusters is defined by **average linkage**:

$$d(A, B) = \frac{1}{|A||B|} \sum_{i \in A} \sum_{j \in B} d(i, j)$$

where $d(i, j)$ is the distance between entries i and j .

- Produces a **dendrogram**, which can be cut at any level to obtain a flat clustering.

Living edition Dendrogram



Structure and Fractures in the Network

- The structure shows a clear divide between contemporary economics and HET clusters.
- A miscellaneous cluster (mostly entries without bibliography, area studies, biographies) floats in between.
- **Surplus approach** (post-Keynesian, Marxian, Sraffian) is more tightly linked with HET than with mainstream clusters.
 - Heterodox see history as critical for progress?
 - Or remains anchored to past traditions, less integrated into current innovation dynamics?
- Neoclassical economics permeates the network but is rarely alone: the **mainstream** is much wider.

Cluster 1-4 (most referenced title by cluster)

C	Label	Title	Authors	#	%
1	General Equilibrium	<i>The Theory of Value</i>	Debreu	24	28.9%
		<i>Markets with a Continuum of Traders</i>	Aumann	20	24.1%
		<i>Elements of Pure Economics</i>	Walras	16	19.3%
2	Quantitative Methods	<i>Foundations of Statistics</i>	Savage	29	13.4%
		<i>Macroeconomics and Reality</i>	Sims	16	7.4%
		<i>Theory of Games and Economic Behavior</i>	Neumann, Morgenstern	16	7.4%
3	Finance	<i>The Pricing of Options and Corporate Liabilities</i>	Black, Scholes	17	27.4%
		<i>Capital Asset Prices</i>	Sharpe	13	21.0%
		<i>A Theory of the Term Structure of Interest Rates</i>	Cox	12	19.4%
4	Monetary & Macro	<i>General Theory</i>	Keynes	24	12.4%
		<i>Time to build and aggregate fluctuations</i>	Kydland, Prescott	20	10.4%
		<i>Interest and Prices</i>	Woodford	19	9.8%

Cluster 5-8

C	Label	Title	Authors	#	%
5	Development & Trade	<i>On the mechanics of economic development</i>	Lucas	14	9.3%
		<i>Increasing returns and long-run growth</i>	Romer	12	7.9%
		<i>The spatial economy</i>	Fujita, Krugman, Venables	12	7.9%
6	Population	<i>A treatise on the Family</i>	Becker	18	10.8%
		<i>On Population</i>	Malthus	17	10.2%
		<i>Economics of discrimination</i>	Becker	12	7.2%
7	Game Theory & Institutions	<i>The problem of social cost</i>	Coase	16	7.4%
		<i>Markets and hierarchies</i>	Williamson	15	6.9%
		<i>The theory of monopolistic competition</i>	Chamberlin	13	6.0%
8	Public	<i>Social Choice and Individual Values</i>	Arrow	22	14.6%
		<i>A pure theory of local expenditures</i>	Tiebout	14	9.3%
		<i>Are government bonds net wealth?</i>	Barro	13	8.6%

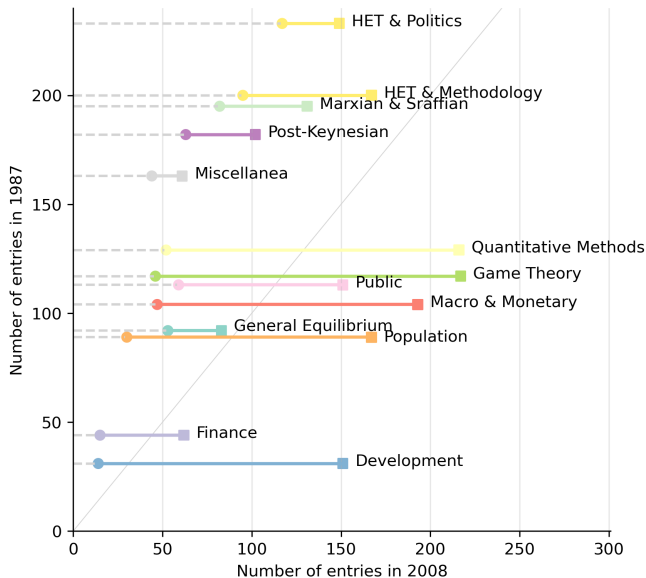
Cluster 10-13

C	Label	Title	Authors	#	%
10	Post-Keynesian	<i>General Theory</i>	Keynes	34	33.3%
		<i>A treatise on money</i>	Keynes	25	24.5%
		<i>Studies in the theory of international trade</i>	Viner	17	16.7%
11	Sraffian & Marxian	<i>Production of Commodities by means of Commodities</i>	Sraffa	47	35.9%
		<i>Capital</i>	Marx	37	28.2%
		<i>Wealth of Nations</i>	Smith	22	16.8%
12	HET & Politics	<i>History of economic analysis</i>	Schumpeter	38	25.5%
		<i>Wealth of Nations</i>	Smith	21	14.1%
		<i>Principles of Political E.</i>	Mill	17	11.4%
13	HET & Methodology	<i>Wealth of Nations</i>	Smith	38	22.8%
		<i>Principles of Economics</i>	Marshall	34	20.4%
		<i>Principles of Political E.</i>	Mill	25	15.0%

Entry counts by cluster across editions

C	Label	1987	2008	Online	Reprints	%87	%08
1	General Equilibrium	92	83	3	53	58%	64%
2	Quantitative Methods	129	216	17	52	40%	24%
3	Finance	44	62	6	15	34%	24%
4	Macro & Monetary	104	193	56	47	45%	24%
5	Development	31	151	38	14	44%	9%
6	Population	89	167	18	30	34%	18%
7	Game Theory	117	217	33	46	39%	21%
8	Public	113	151	22	59	52%	39%
9	Miscellanea	163	61	30	44	27%	72%
10	Post-Keynesian	182	102	21	63	35%	62%
11	Marxian & Sraffian	195	131	11	82	42%	63%
12	HET & Politics	233	149	18	117	50%	79%
13	HET & Methodology	200	167	11	95	47%	57%

Clusters across editions



Topics update

1987 vs. 2008 entry composition by cluster

- Update patterns reveal both epistemic stability and breaks, as well as shifting research frontiers.
- **Highest update intensity:**
 - Development and population studies, especially labor, unemployment, innovation, trade (empirical advancements).
 - Game theory, information, behavioral, experimental.
 - Urban/regional/spatial economics (entirely new).
- **Least updated clusters:**
 - General equilibrium and basic microeconomics (few new contributions, many reprints).
 - Strong cut to surplus approach.

Most referenced titles by edition

Title	Author	# 1987	# 2008	Variation
<i>Capital</i>	Marx	58	26	-55%
<i>Production of Commodities</i>	Sraffa	70	35	-50%
<i>Principles of Political Economy</i>	Mill	56	44	-21%
<i>General Theory</i>	Keynes	84	70	-17%
<i>Wealth of Nations</i>	Smith	101	85	-16%
<i>The Theory of Value</i>	Debreu	48	42	-13%
<i>Principles of Economics</i>	Marshall	80	71	-11%
<i>History of Economic Analysis</i>	Schumpeter	59	56	-5%
<i>Social Choice and Individual Values</i>	Arrow	34	37	9%
<i>Foundations of Economic Analysis</i>	Samuelson	42	51	21%
<i>Essays in Positive Economics</i>	Friedman	20	35	75%

Most referenced authors by edition

Author	1987	2008	% 1987	% 2008	Variation
Sraffa	86	42	5,2%	2,3%	-51,2%
Marx	106	59	6,4%	3,2%	-44,3%
Smith	107	91	6,5 %	4,9%	-15,0%
Schumpeter	118	105	7,1%	5,7%	-11,0%
Hicks	137	125	8,3%	6,8%	-8,8%
Keynes	136	125	8,2%	6,8%	-8,1%
Samuelson	160	184	9,7%	9,9%	15,0%
Arrow	130	173	7,9%	9,4%	33,1%
Friedman	96	140	5,8%	7,6%	45,8%
Becker	48	100	2,9%	5,4%	108,3%
Lucas	51	119	3,1%	6,4%	133,3%

Main conclusion

By combining text embeddings, bibliographic data, and clustering techniques, we reconstructed the latent structure of the New Palgrave.

- **1987** is dominated by **two competing paradigms**: neoclassical and surplus approach.
- **2008** presents a **fragmented plurality** of specialized niches anchored to the **stable core** of the marginalist approach.